HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Produced structured logical workflows demonstrating expert reasoning for complex **technical challenges**, breaking down into step-by-step decision-making processes for AI training.
- Generated high-quality training data for **LLMs**, illustrating effective use of tools and reasoning frameworks by documenting expert problem-solving strategies.
- Critically evaluated **AI-generated reasoning** steps, identifying gaps and errors to refine model behavior for real-world engineering tasks.
- Developed **Python scripts** for automation and data pipeline tasks, processing and transforming data for analytical and reporting purposes.
- Implemented **Java Spring Boot** features and REST APIs, contributing to the development and maintenance of enterprise services.
- Managed production incidents for enterprise applications, performing root cause analysis and ensuring adherence to service level agreements using **Splunk** and **Dynatrace**.
- Designed and implemented **database schema changes** and optimized queries in **PostgreSQL** and **MySQL** to support application requirements.

SKILLS

AI & Machine Learning – Machine Learning Engineer, LLM Training, Reasoning Traces, Prompt Engineering, RLHF Workflows, AI Evaluation Systems, Data Annotation, Structured Problem-Solving

Programming Languages – Python, Java, SQL, R

Frameworks & Libraries – Spring Boot, pandas, NumPy, scikit-learn, Flask, FastAPI

Cloud & DevOps – AWS (EC2, S3, Lambda, CloudWatch), Docker, Jenkins, Git, Linux

Databases – PostgreSQL, MySQL, Oracle

BI & Analytics Tools – Power BI, Tableau, DAX, Power Query

Methodologies – Agile, Scrum, Change Management

PROJECTS

AI Reasoning Trace Generation for Complex Systems

- Developed a framework for generating detailed, step-by-step reasoning traces for complex **technical challenges**, specifically targeting the demonstration of expert thought processes for AI model training.
- Created structured documentation outlining decision-making strategies and problem-solving approaches, serving as high-quality training data to improve **LLM** capabilities in engineering domains.
- Performed manual review and refinement of AI-generated reasoning steps, identifying inconsistencies and errors to enhance the accuracy and logical flow of the training dataset.

Data Pipeline and ML Model for Transaction Analysis

- Engineered a **Python** data pipeline to extract and transform transaction data from diverse sources, preparing a dataset of approximately 150,000 records for analysis.
- Trained a **scikit-learn** classification model on the prepared data to identify patterns indicative of operational anomalies, achieving an AUC of 0.78 on a held-out test set.
- Developed **SQL queries** to validate data integrity and join disparate data streams within the PostgreSQL database prior to model training.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)